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Balancing Novelty and Accuracy in Recommendation

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Balancing Novelty and Accuracy in Recommendation

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A Project Submitted in Partial Fulfillment of the Requirements
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หัวข้อโครงงาน	การสร้างสมดุลระหว่างความแปลกใหม่และความแม่นยำในการแนะนำ
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นายรหัทวิชญ์ ลิขิตบันเทิงพงศ์: การสร้างสมดุลระหว่างความแปลกใหม่และความแม่นยำในการแนะนำ (Balancing Novelty and Accuracy in Recommendation) อ.ที่ปรึกษาโครงงานหลัก : รศ.ดร.ศรันญา มณีโรจน์ 46 หน้า.

ระบบแนะนำมีบทบาทสำคัญอย่างยิ่งในการช่วยให้ผู้ใช้สามารถค้นหาเนื้อหาที่ตรงความต้องการบนแพลตฟอร์มดิจิทัลสมัยใหม่ อย่างไรก็ตาม ระบบส่วนใหญ่ในปัจจุบันมักมุ่งเน้นไปที่ความแม่นยำเป็นหลัก ซึ่งก่อให้เกิดปัญหาที่ผู้ใช้จะได้รับคำแนะนำเพียงข้อมูลที่คุ้นเคยในขอบเขตที่จำกัด และเป็นการปิดกั้นโอกาสในการค้นพบเนื้อหาใหม่ๆ ที่น่าสนใจ วิธีการส่วนใหญ่มักพยายามแก้ไขปัญหานี้โดยใช้กระบวนการกรองภายหลังจากการจัดอันดับเสร็จสิ้นลงแล้ว ทว่าวิธีการดังกล่าวมีข้อจำกัดเนื่องจากรายการคำแนะนำสุดท้ายยังคงถูกจำกัดอยู่ในกลุ่มข้อมูลเริ่มต้นที่คัดเลือกมาตามเกณฑ์ความแม่นยำเพียงอย่างเดียว ซึ่งมักจะละเลยไอเทมในกลุ่ม long-tail หรือเนื้อหาเฉพาะกลุ่ม งานวิจัยนี้จึงนำเสนอ BaNANN ซึ่งเป็นโครงสร้างที่ออกแบบมาเพื่อสร้างสมดุลระหว่างความแม่นยำและความแปลกใหม่โดยตรงในระหว่างกระบวนการเทรนโมเดล โมเดลของเราใช้โครงสร้างกราฟแบบไตรพาร์ไทต์ซึ่งประกอบด้วยผู้ใช้ ไอเทม และประเภทเนื้อหา เพื่อช่วยให้โมเดลสามารถวิเคราะห์บริบทของเนื้อหาได้ดียิ่งขึ้น ด้วยการบูรณาการการเรียนรู้เชิงเปรียบเทียบแบบคู่ โมเดลจะเรียนรู้ที่จะผลักดันผู้ใช้ไปสู่ประเภทเนื้อหาที่ยังไม่เคยสำรวจ ในขณะเดียวกันก็ยังคงรักษาความสอดคล้องทางความหมายและความเกี่ยวข้องของคำแนะนำเอาไว้ ผลการทดสอบบนชุดข้อมูล MovieLens แสดงให้เห็นว่า BaNANN มีประสิทธิภาพเหนือกว่าโมเดลพื้นฐานอย่างมีนัยสำคัญในทุกตัวชี้วัดด้านความแปลกใหม่ ในขณะเดียวกัน ค่าความแม่นยำยังคงรักษาอยู่ในระดับที่สูงและสามารถแข่งขันได้

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RAHATTAWIT LIKITBUNTHENGPONG: BALANCING NOVELTY AND ACCURACY IN RECOMMENDATION.
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Recommender systems are essential for helping users find relevant content on modern digital platforms. However, most traditional systems focus primarily on accuracy, this problem limits users to a narrow range of familiar items and prevents them from discovering new and interesting content. Many current methods try to solve this by using post-processing filters after the ranking is already finished. These methods are limited because the final recommendations are still restricted by the initial accuracy-based candidate pool, which often ignores long-tail or niche items.

We propose BaNANN, a framework designed to balance accuracy and novelty directly during the model training process. Our model uses a tripartite graph structure (users, items, and genres) to tell whether an item is new or not. It uses genre edges to check if an item belongs to a category that the user has not explored before. To achieve this balance, we use the BPR loss function to focus on accuracy and keep recommendations relevant. At the same time, we develop a contrastive loss function to handle the novelty objective by pulling these unexplored genres closer to the user. Experimental results on the MovieLens dataset show that BaNANN performs significantly better than baseline models in all novelty metrics. At the same time, accuracy remains at a high and competitive level.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Recommender systems have become essential parts of modern digital platforms, helping users find relevant content when there's too much information available. From online shopping sites to streaming services and social media, these systems help billions of users navigate through huge collections of products, movies, music, and other digital content. Traditional recommender systems try to maximize accuracy prediction. While these make sure recommendations will match closely with what users have shown they like, they have accidentally created a big problem: the filter bubble effect, where users get stuck in narrow content spaces that only show things similar to what they liked before, limiting their chance to see diverse and potentially interesting content and often ignoring broader goals of recommendation quality, especially the need to introduce users to new content that broadens their interests. This focus only on accuracy can lead to biased recommendations that are statistically correct but fail to help users meaningfully explore and discover new things.

The conflict between accuracy and novelty represents a challenge in recommender system design. Accuracy ensures that users get content they will likely enjoy based on their past behavior, keeping user satisfaction high in the short term. Novelty, on the other hand, introduces users to items or content categories they haven't explored before, potentially improving long-term satisfaction and platform diversity. These goals naturally conflict: the most accurate recommendations are often similar to users' past interactions, while novel recommendations may seem less relevant but offer greater potential for discovery and pleasant surprises.

Current methods that claim to balance accuracy and novelty often use post-processing. The system first produces a ranked list that is completely based on accuracy, then applies a simple filter, for example pushing less popular items higher or dropping the most popular ones down. At first, this may look like the system is adding novelty, but in reality it is not a true trade-off. The reason is that the recommendation list still comes entirely from the original accuracy ranking, so items that never appear in that list, such as niche or long-tail content, will never be recommended. In other words, the system is only reshuffling what it already considers accurate, instead of systematically exploring new options. This makes novelty only a surface adjustment rather than a core part of the model. As a result, users do not get a genuine balance between exploration and exploitation, but rather a slightly modified version of the same accuracy-driven

results, which limits their chance to truly discover new and unexpected content [1].

Recent developments in graph neural networks (GNNs) have greatly improved recommender systems by modeling complex user–item interactions and capturing higher-order collaborative signals. These methods have shown strong performance for accuracy, but they often give little attention to novelty. A key idea in our work is that novelty is easier to measure and manage when recommendations are connected through content categories such as genres. In a pure user–item graph, it is difficult to tell whether an item is "new" for the user, because all items are treated as individual nodes without context. By introducing genres as intermediate nodes, the system can reason at a higher level: it can see which genres a user already knows well and which genres remain underexplored. This approach is inspired by prior work on tripartite graph recommender systems for long-tail items [2], which demonstrated the benefit of an explicit intermediate layer.

In addition to a standard message-passing backbone, we consider *Graph Attention Networks (GAT)* [3] to weight edges in the tripartite user–genre–item graph. Attention allows the model to place higher importance on informative relations (e.g., underexplored genres or genre–item links that best characterize a category), which supports novelty-aware reasoning while preserving accuracy.

Here we presents BaNANN, a novel framework that balances the competing goals of recommendation accuracy and novelty. We integration using a Tripartite Graph Neural Network and Dual Contrastive Learning during the model training process. On the accuracy side, the model employs Bayesian Personalized Ranking (BPR), a widely used objective for implicit feedback. BPR ensures that the system can reliably distinguish which items a user is more likely to prefer, keeping recommendations grounded in the users established tastes. This provides the stable backbone for delivering content that feels relevant and trustworthy. On the novelty side, BaNANN introduces a contrastive learning approach. At the user–genre level, the model makes users closer to genres they have under-explored, creating opportunities for discovery without pushing them into completely unrelated areas. The key is that focusing on a single item often highlights very small, specific details, making the recommendation too narrow. By contrast, working at the genre level helps us push exploration across a wider boundary (a whole group of items) that the user hasn't seen, leading to discovery that feels more natural and meaningful. At the genre–item level, the model strengthens the alignment between genres and their items, so that when users are guided toward new genres, the recommended items are still coherent and meaningful. Together, these contrastive components ensure that novelty emerges as part of the models learned representation, rather than as a surface adjustment afterward.

By combining BPR for accuracy and contrastive learning for novelty, BaNANN learns to provide recommendations that are both familiar and fresh. Users continue to

receive items they are likely to enjoy, while gradually being exposed to content that expands their horizons. This integrated design moves beyond post-processing methods and creates a systematic trade-off between exploration and exploitation.

1.2 Objectives

This paper introduces BaNANN (Balancing Novelty and Accuracy with a Tripartite Graph Neural Network and Dual Contrastive Learning), a framework designed to balance the two conflicting goals of recommender systems: accuracy and novelty by integrating both goals directly into its training process.

1. A **tripartite user–genre–item graph** that makes genre-level reasoning explicit (inspired by extended tripartite graphs [2]).
2. An **accuracy objective (BPR loss)** that keeps rankings consistent with user preferences.
3. A **novelty objective (contrastive loss)** that systematically encourages exploration in under-explored genres and maintains semantic consistency at the item level.

1.3 Scope

Use MovieLens dataset and evaluate accuracy and novelty metrics compared with baseline models, Optional ablations.

1.4 Project Activities

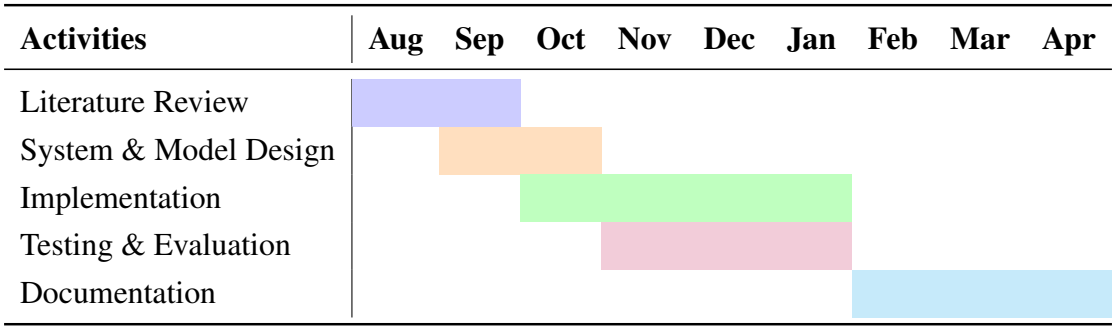
The project will be conducted in sequential phases, aligned with research methodology and system development principles. The main activities are listed below:

1. **Literature Review**
   Study recommender systems, novelty/accuracy trade-offs. Finalize the problem statement and evaluation plan.
2. **System and Model Design**
   Design the tripartite user–genre–item schema; define training objectives (BPR for accuracy, dual contrastive for novelty). Select the backbone and attention strategy (PyTorch Geometric: `GATConv`/heterogeneous GAT).
3. **Implementation**
   Implement BaNANN in PyTorch Geometric with a GAT-based encoder for the tripartite graph; build data preprocessing and evaluation pipelines.

4. **Testing and Evaluation**
   Run experiments on MovieLens; compare to BPR/LightGCN and re-ranking baselines [1]. Report accuracy (Hit@K, NDCG, Precision, Recall, MRR) and novelty metrics.

5. **Documentation and Reporting**
   Prepare reports and the final project report according to academic guidelines.

Figure 1.1: Gantt Timeline (Aug 2025 – Apr 2026)



1.5 Benefits

1. **For the researcher**

   - Gain practical experience in designing and implementing advanced Graph Neural Networks (GNNs).
   - Learn how to apply modern techniques like Contrastive Learning to balance competing objectives.
   - Develop skills in evaluating and analyzing the trade-off between recommendation accuracy and novelty.

2. **For business and society**

   - It helps mitigate the filter bubble effect by providing users with more diverse and novel recommendations and can be applied to business platforms (e.g., e-commerce, streaming) to improve long-term user engagement.

1.6 Report Outline

This report is organized into five chapters as follows:

Chapter I: Introduction Introduces the background, research objectives, and the scope of the proposed BaNANN framework.

Chapter II: Literature Review Reviews related works, existing baseline models, and foundational theories.

Chapter III: Methodology Details the proposed BaNANN architecture.

Chapter IV: Experimental Results presents the experimental setup, evaluation metrics, and a comprehensive analysis of the results using the MovieLens dataset.

Chapter V: Conclusion Summaries of this work.

CHAPTER 2

LITERATURE REVIEW

2.1 Recommender System

Recommender systems are designed to address the information overload problem by filtering large item catalogs and surfacing content that aligns with individual user preferences. While accuracy-focused methods have achieved strong performance in predicting user preferences, optimizing for accuracy alone tends to reinforce existing consumption patterns. Over time, this creates a *filter bubble* effect, where users are repeatedly exposed to a narrow slice of the catalog that closely resembles their past interactions, limiting their opportunity to discover new content . This tension between accuracy and novelty motivates the exploration of recommendation approaches that can balance both objectives.

Balancing accuracy and novelty is a fundamental challenge in recommender system design. On one side, accuracy ensures that users receive content they are likely to enjoy based on their past behavior, maintaining short-term satisfaction. On the other side, novelty exposes users to items or content categories they have not yet explored, which can improve long-term engagement and platform diversity. These two objectives are naturally in tension, as the most accurate recommendations tend to stay close to what the user already knows, leaving little room for discovery.

2.2 Existing Methods for Balancing Accuracy and Novelty

A common approach to introduce novelty is post-processing re-ranking, where an accuracy-oriented model first generates a candidate list, and a secondary step adjusts the ranking to reduce popularity bias or increase diversity [1] [6]. However, this strategy has a fundamental limitation: because the candidate pool is drawn entirely from the accuracy-ranked output, items that never appear in the initial list such as long-tail or genre-diverse content remain invisible regardless of how the re-ranking is performed. Novelty is therefore introduced only as a surface-level adjustment rather than as a core learning objective, and the system cannot achieve genuine content exploration. The following subsections describe two representative methods that illustrate this limitation and serve as baselines in our experiments.

2.2.1 Striking the Balance Between Novelty and Accuracy in Location-Based Recommendation Systems

Agrawal et al. [1] proposed a graph-based framework for recommending Points of Interest (POIs) in Location-Based Social Networks (LBSNs). Their method consists of four main steps. First, users' check-in frequencies are converted into a user–venue affinity matrix. Second, this matrix is transformed into a weighted adjacency matrix where venues are nodes and edges represent how often pairs of venues are positively co-rated by users. Third, a personalized user-profile subgraph is constructed by combining the user's own highly-rated venues with the top-rated venues of their social connections, allowing social relationships to expand the candidate pool. Finally, Shannon entropy is computed for each node in the subgraph, where less popular venues yield higher entropy scores. Recommendations are then generated by selecting venues that satisfy both a minimum occurrence threshold (for accuracy) and a minimum entropy threshold (for novelty), as illustrated in Figure 2.1.

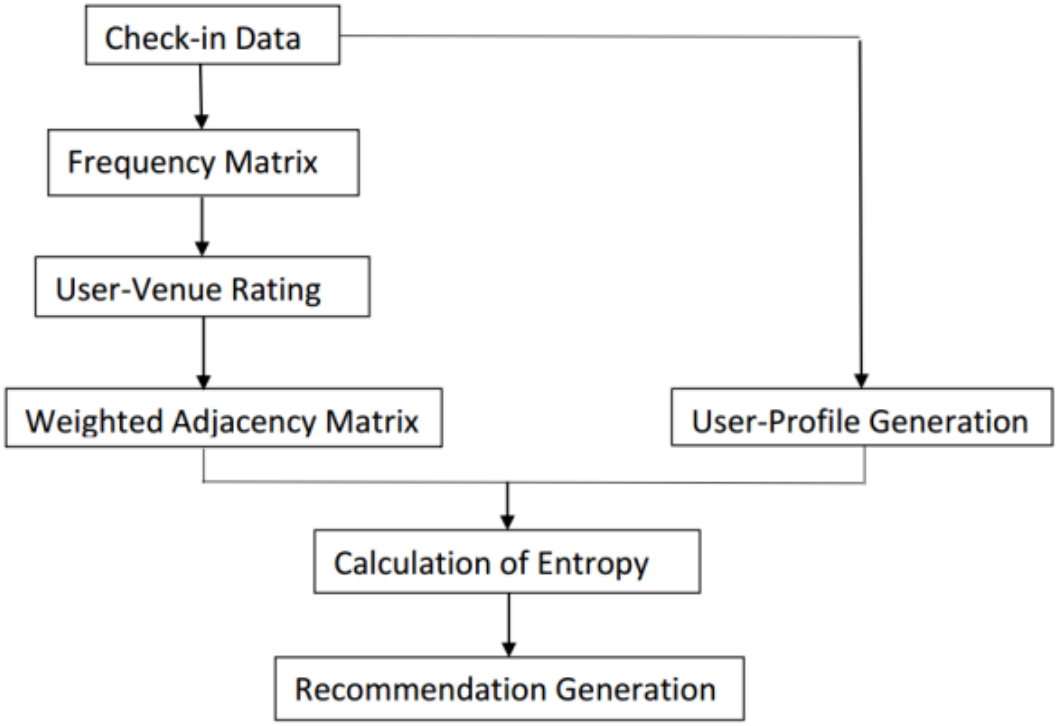


Figure 2.1: Overall framework of the model proposed by Agrawal et al. [1] for balancing novelty and accuracy in location-based recommendation.

However, novelty in this framework is introduced only as a post-processing filter applied after the candidate list has already been formed from accuracy-driven signals. Items that do not appear in the initial graph are never considered, meaning that the depth of exploration is fundamentally bounded by what the accuracy stage already surfaced. Novelty is therefore a surface adjustment rather than a core learning objective.

2.2.2 Diversity and Serendipity Preference-Aware Recommender System

Yin and Zhao [6] proposed DSPA-RS, a preference-aware framework that simultaneously optimizes two complementary objectives: diversity preference and serendipity preference. Their key observation is that different users have different tolerances for novel content, and applying a uniform novelty strategy across all users inevitably degrades accuracy for those who prefer familiar recommendations. DSPA-RS addresses this by modeling each user individually. Diversity preference is represented as a vector capturing the distribution of item feature values (e.g., genre, director) in the user's highly-rated history, reflecting what familiar content mix the user expects. Serendipity preference is captured through a novelty momentum vector, which tracks item features that have recently appeared in the user's profile for the first time, weighted by a time-decay parameter. The final recommendation is produced by selecting k items from an accuracy-generated candidate set that maximizes a weighted sum-of-ratios objective combining both preference components, with a personalized balance parameter α^i controlling the relative importance of familiarity versus novelty for each user.

Nevertheless, DSPA-RS still operates as a post-processing re-ranking step over a candidate list generated by an accuracy-oriented model (NCF). Because novelty is introduced only after the candidate pool has been fixed, items outside that pool are never reached regardless of how well a user's novelty momentum is estimated. As a result, genuine genre-level exploration beyond the user's familiar content boundary remains out of reach.

2.3 Graph Neural Network

To build upon the concepts of modeling complex relationships, Graph Neural Networks (GNNs) have greatly improved the performance of recommender systems [5]. In standard graph-based recommender systems, the graph only has two types of nodes: users and items (bipartite graph). However, to help the system understand categories and manage novelty, a tripartite graph architecture is used. This graph has three types of nodes: users, items, and genres. Adding genre nodes helps the system understand which categories the user knows well and which ones are new to them.

Because this graph has different types of nodes and edges, it requires a Heterogeneous Graph Neural Network (Hetero-GNN). The message-passing process for each edge type (s,r,t), which includes a source node s, a target node t, and a relation type r, is calculated as:

$$h_{t,v}^{(l+1)} = \sigma\left(W_r^{(l)} \cdot \mathrm{AGG}_{u \in N_r(v)}(h_{s,u}^{(l)})\right) \tag{2.1}$$

where $h_{t,v}^{(l)}$ is the embedding of node v of type t at layer l, $N_r(v)$ is the set of neighbors connected to v by relation r, $W_r^{(l)}$ is a weight matrix, and the aggregation function AGG is computed using mean pooling.

2.4 Graph Attention Network (GAT)

Standard GNNs treat all neighboring nodes equally by assigning uniform weights during aggregation. Graph Attention Networks (GAT) [4] address this limitation by introducing a learnable attention mechanism that computes a dynamic importance score for each edge, allowing the model to focus on the most informative neighbors.

For a source node u and a target node v connected by an edge, the unnormalized attention coefficient is computed as:

$$e_{uv} = \text{LeakyReLU}\left(\mathbf{a}^T \left[\mathbf{W}h_u \| \mathbf{W}h_v\right]\right) \tag{2.2}$$

where $\mathbf{W}$ is a shared linear transformation, $\mathbf{a}$ is a learnable attention vector, and $\|$ denotes concatenation. These coefficients are then normalized across all neighbors of v using a softmax:

$$\alpha_{uv} = \frac{\exp(e_{uv})}{\sum_{k \in \mathcal{N}(v)} \exp(e_{kv})} \tag{2.3}$$

The updated embedding of node v is then computed as a weighted sum of its neighbors' representations:

$$h'_v = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{uv} \mathbf{W} h_u\right) \tag{2.4}$$

To stabilize learning, GAT employs *multi-head attention*, where K independent attention heads are applied in parallel and their outputs are concatenated, as illustrated in Figure 2.2:

$$h'_v = \|_{k=1}^{K} \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{uv}^{(k)} \mathbf{W}^{(k)} h_u\right) \tag{2.5}$$

As shown in Figure 2.2, the left side illustrates how attention coefficients α_{uv} are computed between a target node and its neighbors, with line thickness indicating attention weight. The right side shows multi-head attention, where outputs from K independent heads are concatenated to form the final node representation.

In our tripartite graph, GAT is applied at each heterogeneous edge type. The attention mechanism is particularly suited to this setting because different edges carry different levels of information depending on the propagation direction—for example, weighting the *genre* $\rightarrow$ *item* edges to reflect how well an item represents its genre, or weighting *user* $\rightarrow$ *item* edges to capture strong historical preferences. Rather than guiding the model explicitly toward unexplored genres through attention weights, novelty

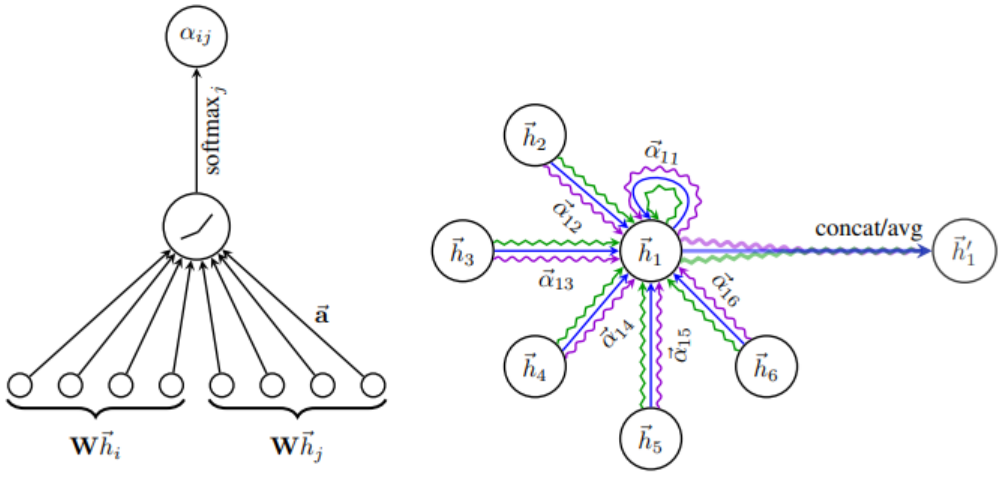


Figure 2.2: The mechanism of a Graph Attention Network (GAT).

is introduced through the contrastive learning objectives described in Chapter 3, while GAT provides an expressive and adaptive aggregation backbone that supports both objectives.

2.5 Bayesian Personalized Ranking (BPR)

To learn what users like from implicit feedback (like clicks or views), Bayesian Personalized Ranking (BPR) is a popular method [3]. BPR is a pairwise ranking loss function. Its main idea is that a user prefers an item they have clicked on over an item they have never seen.

The preference score between a user u and an item i is calculated using their embedding vectors:

$$s(u,i) = e_u^T e_i \tag{2.6}$$

where e_u and e_i are the embedding vectors of the user and the item.

The BPR loss function is defined as:

$$\mathscr{L}_{acc} = -\sum_{(u,i,j)\in D_s} \log \sigma(s(u,i) - s(u,j)) \tag{2.7}$$

where D_s is the training set containing a user u, a clicked item i (positive sample), and an unclicked item j (negative sample), and $\sigma(\cdot)$ is the sigmoid function. By minimizing this loss, the model learns to rank the items that users like higher than the unseen items, helping the system provide accurate recommendations.

2.6 Contrastive Learning

Contrastive learning is a self-supervised learning method that improves embeddings. Its main goal is to pull similar data (positive pairs) closer together and push different data (negative pairs) apart in the embedding space. To achieve this, the InfoNCE (Information Noise-Contrastive Estimation) loss function is widely used. InfoNCE helps the model identify the correct positive sample from a large set of negative samples.

The general formula for the InfoNCE loss is defined as:

$$\mathscr{L}_{InfoNCE} = -\log \frac{\exp(s(u, v^+)/\tau)}{\sum_{v \in V} \exp(s(u, v)/\tau)} \tag{2.8}$$

where $s(\cdot)$ is a scoring function, such as a dot product, that measures the similarity between two embeddings. Here, u is the target node, v^+ is the positive sample, and V is a set containing both the positive sample and many negative samples. The parameter τ is the temperature, which controls how much the model penalizes hard negative samples.

By minimizing the InfoNCE loss, the model tries to maximize the similarity of the positive pair in the numerator while minimizing the similarities of all negative pairs in the denominator. This process maximizes the mutual information between the positive pairs.

In our work, InfoNCE is used to manage recommendation novelty. By creating special contrastive tasks based on the InfoNCE framework such as pulling user embeddings closer to their unexplored genre embeddings or making sure genres and their items are similar. This helps the model learn to balance accuracy and novelty directly during training, which is much more effective than using traditional post-processing methods. More details are described in Chapter 3.

CHAPTER 3

METHODOLOGY

3.1 Overall framework

As we talk, the core challenge in recommender systems lies in balancing the conflicting objectives of accuracy and novelty. Our method addresses this conflict by integrates a tripartite graph architecture, BPR, and two-step contrastive learning for balanced objective optimization. Our framework is show in figure 2.

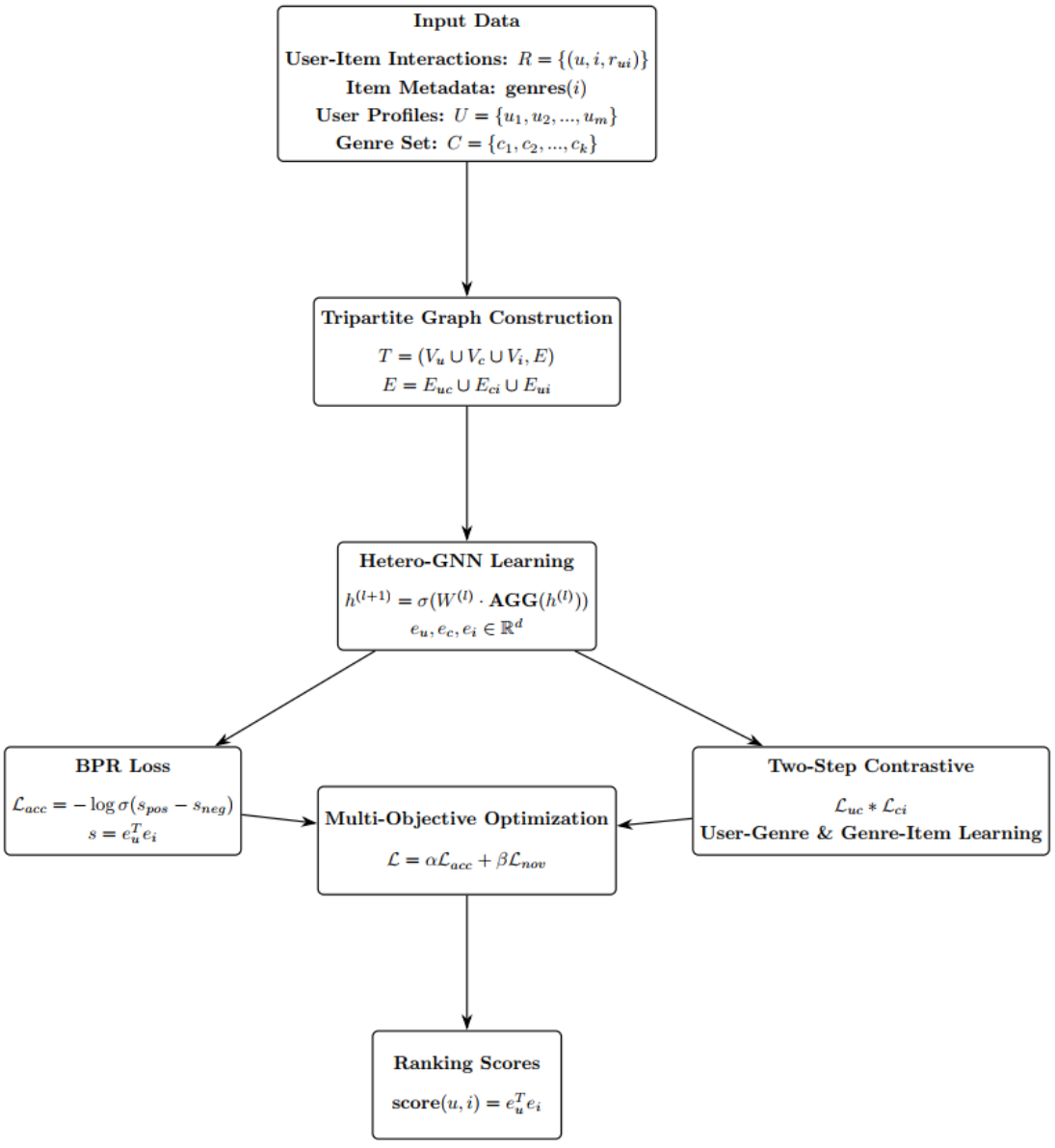


Figure 3.1: our model framework

3.1.1 Tripartite Graph Construction

Let $U = \{u_1, u_2, \ldots, u_m\}$ denote the set of m users, $I = \{i_1, i_2, \ldots, i_n\}$ the set of n items, and $C = \{c_1, c_2, \ldots, c_k\}$ the set of k genres. Observed user-item ratings are represented as $R = \{(u, i, r_{ui}) | u \in U, i \in I\}$, where r_{ui} denotes the ratings.

Tripartite Graph Construction

We propose a tripartite graph structure that explicitly incorporates content semantics through genre nodes, creating a three-layer architecture:

$$V = V_u \cup V_c \cup V_i \tag{3.1}$$

where $V_u = \{v_{u1}, \ldots, v_{um}\}$ are user nodes, $V_c = \{v_{c1}, \ldots, v_{ck}\}$ are genre nodes, and $V_i = \{v_{i1}, \ldots, v_{in}\}$ are item nodes.

The edge set E encompasses three distinct relationship types:

User-Item Edges (E_{ui}): Store direct collaborative filtering signals essential for accuracy. For each observed interaction $(u, i) \in R$, we create an edge if $r_{ui} \in R$.

Genre-Item Edges (E_{ci}): Encode content-based relationships that enable semantic reasoning. An edge $e_{ci} \in E_{ci}$ exists if item i belongs to genre c, based on item metadata.

User-Genre Edges (E_{uc}): Capture implicit user preferences at the content category level. For each user-genre pair (u, c), we first identify all items that user u has interacted with belonging to genre c:

$$I_{uc} = \{i \in I | (u, i) \in R \text{ and } (c, i) \in E_{ci}\} \tag{3.2}$$

The edge weight w_{uc} is computed as the average interaction strength across these items:

$$w_{uc} = \frac{1}{|I_{uc}|} \sum_{i \in I_{uc}} r_{ui} \tag{3.3}$$

Information Flow and Multi-hop Propagation

The tripartite structure enables rich information propagation patterns that enhance both accuracy and novelty:

One-hop Paths (User $\leftrightarrow$ Item/User $\leftrightarrow$ Genre): Preserve traditional collaborative filtering signals.

Two-hop Paths (User $\leftrightarrow$ Genre $\leftrightarrow$ Item/User $\leftrightarrow$ Item $\leftrightarrow$ Genre): Enable content-based reasoning and genre-level preference learning.

Note that our tripartite graph contains six directed edge types: $(user, rates, item)$ and $(item, ratesBy, user)$; $(user, likes, genre)$ and $(genre, likesBy, user)$; $(genre, has, item)$

and $(item, isIn, genre)$.

3.1.2 Heterogeneous Graph Neural Network Learning

Our tripartite graph contains multiple node types and edge types, requiring a heterogeneous graph neural network architecture. We employ HeteroConv layers that apply different transformations to different edge types while enabling information flow across the entire graph.

For each edge type (s,r,t) where s and t are source and target node types and r is the relation type, we define a relation-specific convolution:

$$h_{t,v}^{(l+1)} = \sigma\left(W_r^{(l)} \cdot \mathrm{AGG}_{u \in N_r(v)}\left(h_{s,u}^{(l)}\right)\right) \tag{3.4}$$

where $h_{t,v}^{(l)}$ is the embedding of node v of type t at layer l, $N_r(v)$ represents neighbors of v connected via relation r, and $W_r^{(l)}$ is a weight matrix.

We implement the aggregation function using mean pooling:

$$\mathrm{AGG}_{u \in N_r(v)}\left(h_{s,u}^{(l)}\right) = \mathrm{MEAN}\left\{h_{s,u}^{(l)} : u \in N_r(v)\right\} \tag{3.5}$$

3.1.3 BPR Learning

We use a simple BPR loss show as follows:

$$s(u,i) = e_u^T e_i \tag{3.6}$$

For each user u, observed item i, and unobserved item j, the loss of BPR is the following.

$$\mathscr{L}_{\mathrm{acc}} = -\sum_{(u,i,j) \in D_s} \ln \sigma(s(u,i) - s(u,j)) \tag{3.7}$$

where D_s is the training set and $\sigma(\cdot)$ is the sigmoid function.

3.1.4 Two-Step Contrastive Learning Framework

User-Genre Contrastive Learning

For each user u, we analyze their interaction pattern across genres to compute a preference distribution. Let n_{uc} denote the number of interactions user u has had with items in genre g:

$$n_{uc} = \sum_{i \in I:(c,i) \in E_{ci}} \mathbb{I}[(u,i) \in R] \tag{3.8}$$

where the indicator function $\mathbb{I}[A]$ is defined as:

$$\mathbb{I}[A] = \begin{cases} 1 & \text{if condition } A \text{ is true} \\ 0 & \text{otherwise} \end{cases} \tag{3.9}$$

The probability that user u interacts with genre c is:

$$p_{uc} = \frac{n_{uc}}{\sum_{c' \in C} n_{uc'}} \tag{3.10}$$

The first step is to encourage users to be attracted to genres they haven't sufficiently explored. So, for each user u, we identify unexplored genres as positive samples. A genre c is considered unexplored for user u if $p_{uc} < \varepsilon$ where $\varepsilon = 1/|C|$ represents uniform exploration.

Let C_u^- denote the set of unexplored genres for user u:

$$C_u^- = \{c \in C | p_{uc} < \varepsilon\} \tag{3.11}$$

The user-genre contrastive loss encourages user embeddings to be similar to unexplored genre embeddings:

$$\mathscr{L}_{uc} = -\sum_{u \in U} \sum_{c \in C_u^-} \ln \frac{\exp(e_u^T e_c / \tau)}{\sum_{c' \in C} \exp(e_u^T e_{c'} / \tau)} \tag{3.12}$$

Genre-Item Contrastive Learning

The second step ensures that when users are attracted to new genres, the recommended items within those genres are semantically appropriate. For each genre c, we want item embeddings within that genre to be similar to the genre embedding.

Let I_c denote the set of items belonging to genre c:

$$I_c = \{i \in I | (c,i) \in E_{ci}\} \tag{3.13}$$

The genre-item contrastive loss is:

$$\mathscr{L}_{ci} = -\sum_{c \in C} \sum_{i \in I_c} \ln \frac{\exp(e_c^T e_i / \tau)}{\sum_{i' \in I} \exp(e_c^T e_{i'} / \tau)} \tag{3.14}$$

3.1.5 Multi-Objective Optimization

The overall novelty loss combines both contrastive components:

$$\mathscr{L}_{\text{nov}} = \mathscr{L}_{uc} \mathscr{L}_{ci} \tag{3.15}$$

Then we combine BPR loss and contrastive loss into the final loss function as follows:

$$\mathscr{L} = \alpha \mathscr{L}_{\text{acc}} + \beta \mathscr{L}_{\text{nov}} \tag{3.16}$$

3.1.6 Inference and Recommendation Generation

During inference, we compute recommendation scores using the simple scoring function used during training:

$$\text{score}(u,i) = e_u^T e_i \tag{3.17}$$

Our task is to perform ranking. After this score is computed for all candidate items i for a given user u, we generate the final top-K recommendation list by ranking these items in descending order of their scores.

CHAPTER 4

EXPERIMENTAL RESULTS

In this chapter, we evaluate the performance of our proposed BaNANN framework. We first describe the dataset used in our experiments, followed by the baseline models and the evaluation metrics for both accuracy and novelty. We then detail the implementation settings. Finally, we present the performance comparison and discuss the effectiveness of our approach in balancing accuracy and novelty.

4.1 Experimental Settings

4.1.1 Dataset

To evaluate the effectiveness of our proposed model, we conduct experiments on the widely used MovieLens dataset, specifically the **MovieLens 100K (ml-latest-small)** version. This dataset contains 100,836 ratings and 3,683 tag applications across 9,742 movies, created by 610 users. The presence of explicit movie genre information makes it highly suitable for evaluating our tripartite graph architecture (User-Item-Genre). To adapt the explicit 5-star ratings for our implicit feedback task (BPR optimization), we treat observed ratings as positive feedback.

4.1.2 Baseline Models

To demonstrate the superiority of our proposed model in balancing multiple objectives, we compare it against the following representative baseline models, including the existing methods reviewed in Chapter 2 and an ablation variant of our model:

- **GAT+BPR (Ablation Baseline)**: A variant of our proposed model that open only the BPR loss and close the contrastive learning part. It serves to demonstrate the pure accuracy focus of the architecture.
- **DSPA-RS**: Diversity and Serendipity Preference-Aware Recommender System [6]. A baseline model focusing on beyond-accuracy metrics to customize the level of novelty for users.
- **SBNA-L**: Striking the Balance Between Novelty and Accuracy in Location-Based Recommendation System [1]. A baseline approach that attempts to balance relevance and exploration using graph-based relationships.

4.1.3 Leave-One-Out Evaluation

To evaluate the performance of the recommender system, we adopt the Leave-One-Out (LOO). For each user $u \in U$, we keep the last interaction (the most recent item the user interacted with) as the ground-truth test item i_{test}. The remaining interactions for each user are used to construct the training set R_{train}. This approach simulates a real-world scenario where the system predicts the next item a user will likely interact with based on their complete past behavior.

During the evaluation phase, to make the ranking task more challenging and computationally efficient, we follow a **Sampling Strategy** for each user as follows:

1. **Positive Sample:** The positive sample is the single held-out test item i_{test}.
2. **Negative Samples:** We randomly sample 99 items from the set of items that the user has never interacted with in the training, validation, or test sets ($I \setminus R_{u,all}$).
3. **Ranking Task:** These 100 items (1 positive and 99 negatives) form a candidate list. The model computes a preference score for each item in the list, and we then rank them in descending order to calculate the accuracy and novelty metrics.

4.1.4 Evaluation Metrics

We evaluate the Top-K recommendation performance using the following metrics under the Leave-One-Out protocol, where each user has exactly one ground-truth test item i_{test}.

Accuracy Metrics

- **Precision@K:** The proportion of relevant items in the Top-K from top-K recommended list L_u. (Since there is exactly one ground-truth item per user under LOO, this can be simplifies to Precision@K = HR@K / K. While HR means Hit-rate.)

$$\text{Precision@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|L_u \cap \{i_{test}\}|}{K} \tag{4.1}$$

- **Recall@K:** The proportion of ground-truth items retrieved in the Top-K list. Under LOO, Recall@K is identical to HR@K since there is only one ground-truth item per user.

$$\text{Recall@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|L_u \cap \{i_{test}\}|}{|\{i_{test}\}|} \tag{4.2}$$

- **F1-Score@K:** The harmonic mean of Precision and Recall.

$$\text{F1@}K = \frac{2 \cdot \text{Precision@}K \cdot \text{Recall@}K}{\text{Precision@}K + \text{Recall@}K} \tag{4.3}$$

Note that under LOO, F1@K simplifies to $2/(K+1)$ when the test item is retrieved, and 0 otherwise. It is reported for completeness but is less informative than NDCG or MRR in this setting.

- **NDCG@K:** Normalized Discounted Cumulative Gain, which rewards higher-ranked hits more than lower-ranked ones.

$$\text{NDCG@}K = \frac{1}{|U|} \sum_{u \in U} \frac{\log 2}{\log(\text{rank}_{u,i_{test}} + 1)} \tag{4.4}$$

where $\text{rank}_{u,i_{test}}$ is the position of i_{test} in L_u, and the score is 0 if $i_{test} \notin L_u$.

- **Mean Reciprocal Rank (MRR):** The average of the reciprocal rank of the test item across all users.

$$\text{MRR} = \frac{1}{|U|} \sum_{u \in U} \frac{1}{\text{rank}_{u,i_{test}}} \tag{4.5}$$

where the reciprocal rank is 0 if $i_{test} \notin L_u$.

Novelty and Coverage Metrics

- **AVG Novelty@K:** Measures the unexpectedness of recommended items based on their global popularity, normalized to the $[0,1]$ range. Items that are rarely interacted with yield higher novelty scores.

$$\text{Novelty@}K = \frac{1}{|U|} \sum_{u \in U} \frac{1}{K} \sum_{j \in L_u} \frac{-\log_2\left(\frac{\text{freq}(j)+1}{Z+|I|}\right)}{\log_2(Z+|I|)} \tag{4.6}$$

where $\text{freq}(j)$ is the number of interactions item j has in the training set, $Z = \sum_j \text{freq}(j)$ is the total number of training interactions, and $|I|$ is the number of items in the catalog. Additive smoothing $(\text{freq}(j)+1)$ prevents zero probability for items with no training interactions, and the denominator $\log_2(Z+|I|)$ normalizes the score to $[0,1]$.

- **Unseen-Genre Ratio (UGR@K):** The proportion of recommended items that belong to genres the user has not sufficiently explored, measuring how effectively the model pushes users beyond their existing content boundaries.

$$\text{UGR@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|\{j \in L_u \mid \text{genres}(j) \cap C_u^- \neq \emptyset\}|}{K} \tag{4.7}$$

where C_u^- is the set of unexplored genres for user u as defined in Section 3.4.1, and $\text{genres}(j)$ is the set of genres item j belongs to. An item is counted if it belongs to *at least one* unexplored genre.

- **Coverage@K:** The percentage of all catalog items that are recommended to at least one user, reflecting the system's ability to surface a diverse range of items rather than concentrating on a small popular subset.

$$\text{Coverage@}K = \frac{|\bigcup_{u \in U} L_{u,K}|}{|I|} \tag{4.8}$$

- **Long-tail Share@K:** The proportion of recommendations that belong to the bottom 10% least popular items in the training set, measuring how well the model recommends niche content beyond the popularity bias.

$$\text{Long-tail Share@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|\{j \in L_u \mid \text{freq}(j) \leq \delta_{0.10}\}|}{K} \tag{4.9}$$

where $\delta_{0.10}$ is the 10th percentile of item interaction counts in the training set.

4.1.5 Implementation Details

Our proposed model is implemented using PyTorch and PyTorch Geometric. For the training process, we utilize a Pairwise Sampling strategy to optimize the BPR loss. For each observed interaction (u,i) in the training set (positive sample), we randomly sample one item j from the set of items that user u has never interacted with $(I \setminus R_u)$ to serve as a negative sample. This ensures the model learns to rank preferred items higher than unseen ones.

During the evaluation phase, to efficiently compute metrics, we follow this Sampled Metrics approach. For each user's test item, we randomly select 99 negative items that the user has not interacted with, forming a candidate set of 100 items. The model then ranks these 100 items, and we calculate the metrics based on the position of the ground-truth test item within this ranked list.

4.2 Performance Comparison

To provide a comprehensive evaluation, we present the results separated into two tables. Table 4.1 details the accuracy metrics, and Table 4.2 details the novelty metrics.

For each baseline model, the table presents two rows: the absolute metric value on the top, and the relative percentage improvement **achieved by our model (BaNANN) over that specific baseline** on the bottom. Positive improvements (where our model performs better) are highlighted in blue, while negative differences are marked in red. Bold values indicate the best performance, and underlined values represent the second best.

Table 4.1: Accuracy performance of the proposed model and baselines.

Model	Recall			Precision			F1-score			NDCG			MRR		
	@5	@10	@20	@5	@10	@20	@5	@10	@20	@5	@10	@20	@5	@10	@20
GAT+BPR	**0.5419** (-1.83%)	**0.6765** (-0.96%)	**0.8079** (-3.66%)	**0.1084** (-1.85%)	**0.0677** (-1.03%)	**0.0404** (-3.71%)	**0.1806** (-1.83%)	**0.1230** (-0.98%)	**0.0769** (-3.64%)	0.3958 (+0.23%)	0.4389 (+0.57%)	**0.4743** (-1.20%)	0.3500 (+0.57%)	0.3670 (+0.95%)	0.3773 (+0.16%)
DSPA-RS	0.5279 (+0.78%)	0.6672 (+0.42%)	0.7836 (-0.68%)	0.1056 (+0.76%)	0.0667 (+0.45%)	0.0392 (-0.77%)	0.1760 (+0.74%)	0.1213 (+0.41%)	0.0746 (-0.67%)	0.3693 (+7.42%)	0.4154 (+6.26%)	0.4446 (+5.40%)	0.3169 (+11.08%)	0.3363 (+10.17%)	0.3444 (+9.73%)
SBNA-L	0.4049 (+31.39%)	0.5377 (+24.60%)	0.7033 (+10.66%)	0.0810 (+31.36%)	0.0538 (+24.54%)	0.0352 (+10.51%)	0.1350 (+31.33%)	0.0978 (+24.54%)	0.0670 (+10.60%)	0.2796 (+41.88%)	0.3228 (+36.74%)	0.3643 (+28.63%)	0.2384 (+47.65%)	0.2564 (+44.50%)	0.2675 (+41.27%)
BaNANN (Ours)	0.5320	0.6700	0.7783	0.1064	0.0670	0.0389	0.1773	0.1218	0.0741	**0.3967**	**0.4414**	0.4686	**0.3520**	**0.3705**	**0.3779**

Table 4.2: Novelty performance of the proposed model and baselines.

Model	AVG Novelty				Unseen-Genre Ratio				Coverage				Long-tail Share (10%)			
	@5	@10	@20	@50	@5	@10	@20	@50	@5	@10	@20	@50	@5	@10	@20	@50
GAT+BPR	0.5992 (+19.29%)	0.6234 (+19.63%)	0.6570 (+18.86%)	0.7135 (+15.98%)	0.3038 (+8.33%)	0.3071 (+8.27%)	0.3187 (+8.72%)	0.3300 (+5.88%)	0.1821 (+9.56%)	0.2993 (+14.43%)	0.4834 (+17.46%)	0.8378 (+8.02%)	0.0207 (+195.17%)	0.0368 (+173.64%)	0.0722 (+108.17%)	0.1731 (+44.02%)
DSPA-RS	0.6699 (+6.70%)	0.6998 (+6.57%)	0.7371 (+5.94%)	0.8011 (+3.30%)	0.0689 (+377.65%)	0.0725 (+358.62%)	0.0807 (+329.37%)	0.0850 (+311.06%)	0.1437 (+38.83%)	0.2412 (+42.00%)	0.3925 (+44.66%)	0.6574 (+37.66%)	0.0059 (+935.59%)	0.0121 (+732.23%)	0.0226 (+565.04%)	0.0892 (+179.48%)
SBNA-L	0.6345 (+12.66%)	0.6679 (+11.66%)	0.7138 (+9.40%)	0.7961 (+3.94%)	0.0885 (+271.86%)	0.0869 (+282.62%)	0.0860 (+302.91%)	0.0909 (+284.38%)	0.0833 (+139.50%)	0.1434 (+138.84%)	0.2546 (+123.02%)	0.5620 (+61.03%)	0.0000 (+∞%)	0.0000 (+∞%)	0.0150 (+902.00%)	0.1297 (+92.21%)
BaNANN (Ours)	**0.7148**	**0.7458**	**0.7809**	**0.8275**	**0.3291**	**0.3325**	**0.3465**	**0.3494**	**0.1995**	**0.3425**	**0.5678**	**0.9050**	**0.0611**	**0.1007**	**0.1503**	**0.2493**

4.2.1 Results in Accuracy

The experimental results in Table 4.1 show that the GAT+BPR (Ablation Baseline) achieves the highest performance in general retrieval metrics, such as Recall and Precision. This is because the ablation model focuses exclusively on accuracy optimization without the need to distribute loss weights toward novelty objectives during the training process.

However, both BaNANN and GAT+BPR significantly outperform the other baselines, DSPA-RS and SBNA-L. This superiority stems from the use of Graph Attention Networks (GAT) as the core architecture, which is a more modern and expressive GNN compared to the traditional models used in the baselines.

4.2.2 Results in Novelty

In terms of novelty, BaNANN demonstrates clear superiority, outperforming all baseline models across every metric. The primary reason for this success is that BaNANN integrates exploration goals directly into the model training phase (in-training balance) through the Dual Contrastive Learning framework.

This approach fundamentally differs from baselines like SBNA-L and DSPA-RS, which rely on post-processing re-ranking methods. Such traditional methods are inherently limited because the recommended items are restricted to a candidate pool initially selected based on accuracy, making it difficult to surface genuine long-tail items or unexplored genres. In contrast, BaNANNs training process allows the model to learn balanced representations from the start, leading to significantly higher Unseen-Genre Ratios and Long-tail Shares compared to methods that only apply surface-level adjustments.

CHAPTER 5

CONCLUSION

This research has presented BaNANN, a framework designed to bridge the gap between recommendation accuracy and novelty through a tripartite graph-based architecture and self-supervised contrastive learning. The study was motivated by the limitations of traditional recommender systems, which often prioritize accuracy to the extent of creating a "filter bubble," thereby restricting users to familiar content and neglecting long-tail items. Unlike existing methods that rely on post-processing or re-ranking, BaNANN integrates exploration objectives directly into the model's training phase.

Through the implementation of a tripartite graph consisting of users, items, and genres, the model successfully performed high-level reasoning across content categories. The integration of Graph Attention Networks (GAT) allowed for dynamic weighing of relationships, ensuring that established preferences were maintained while unexplored genres were intelligently prioritized. Furthermore, the dual-step contrastive learning frameworkcomprising User-Genre and Genre-Item tasksensured that the exploration was both systematic and semantically grounded.

The experimental results on the MovieLens 100K dataset, evaluated under the strict Leave-One-Out protocol, demonstrated that BaNANN effectively resolves the accuracy-novelty trade-off. While baseline models like GAT+BPR achieved high accuracy but lacked novelty, and beyond-accuracy baselines such as DSPA-RS and SBNA-L suffered from significant losses in relevance, BaNANN maintained competitive ranking performance while achieving superior scores in novelty, coverage, and long-tail share. Notably, the model's ability to outperform pure accuracy baselines in Mean Reciprocal Rank (MRR) and NDCG at tighter thresholds proves that a well-structured exploration of new content can enhance overall recommendation quality.

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Appendix I

The Project Proposal of Course 2301399
Project Proposal
Academic Year 2025

Project Title (Thai)	
Project Title (English)	Balancing Novelty and Accuracy in Recommendation
Project Advisor	Assoc. Prof. Saranya Maneeroj
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1 Background and Rationale

Recommender systems have become essential parts of modern digital platforms, helping users find relevant content when there's too much information available. From online shopping sites to streaming services and social media, these systems help billions of users navigate through huge collections of products, movies, music, and other digital content. Traditional recommender systems try to maximize accuracy prediction. While these make sure recommendations will match closely with what users have shown they like, they have accidentally created a big problem: the filter bubble effect, where users get stuck in narrow content spaces that only show things similar to what they liked before, limiting their chance to see diverse and potentially interesting content and often ignoring broader goals of recommendation quality, especially the need to introduce users to new content that broadens their interests. This focus only on accuracy can lead to biased recommendations that are statistically correct but fail to help users meaningfully explore and discover new things.

The conflict between accuracy and novelty represents a challenge in recommender system design. Accuracy ensures that users get content they will likely enjoy based on their past behavior, keeping user satisfaction high in the short term. Novelty, on the other hand, introduces users to items or content categories they haven't explored before, potentially improving long-term satisfaction and platform diversity. These goals naturally conflict: the most accurate recommendations are often similar to users' past interactions, while novel recommendations may seem less relevant but offer greater potential for discovery and pleasant surprises.

Current methods that claim to balance accuracy and novelty often use post-processing. The system first produces a ranked list that is completely based on accuracy, then applies a simple filter, for example pushing less popular items higher or dropping the most popular ones down. At first, this may look like the system is adding novelty, but in reality it is not a true trade-off. The reason is that the recommendation list still comes entirely from the original accuracy ranking, so items that never appear in that list, such as niche or long-tail content, will never be recommended. In other words, the system is only reshuffling what it already considers accurate, instead of systematically exploring new options. This makes novelty only a surface adjustment rather than a core part of the model. As a result, users do not get a genuine balance between exploration and exploitation, but rather a slightly modified version of the same accuracy-driven results, which limits their chance to truly discover new and unexpected content [1].

Recent developments in graph neural networks (GNNs) have greatly improved recommender systems by modeling complex user–item interactions and capturing higher-order collaborative signals. These methods have shown strong performance for accuracy, but they often give little attention to novelty. A key idea in our work is that novelty is easier to measure and manage when recommendations are connected through content categories such as genres. In a pure user–item graph, it is difficult to tell whether an item is "new" for the user, because all items are treated as individual nodes without context. By introducing genres as intermediate nodes, the system can reason at a higher level: it can see which genres a user already knows well and which genres remain underexplored. This approach is inspired by prior work on tripartite graph recommender systems for long-tail items [2], which demonstrated the benefit of an explicit intermediate layer.

In addition to a standard message-passing backbone, we consider *Graph Attention Networks (GAT)* [3] to weight edges in the tripartite user–genre–item graph. Attention allows the model to place higher importance on informative relations (e.g., under-explored genres or genre–item links that best characterize a category), which supports novelty-aware reasoning while preserving accuracy.

Here we presents BaNANN, a novel framework that balances the competing goals of recommendation accuracy and novelty. We integration using a Tripartite Graph Neural Network and Dual Contrastive Learning during the model training process. On the accuracy side, the model employs Bayesian Personalized Ranking (BPR), a widely used objective for implicit feedback. BPR ensures that the system can reliably distinguish which items a user is more likely to prefer, keeping recommendations grounded in the users established tastes. This provides the stable backbone for delivering content that feels relevant and trustworthy. On the novelty side, BaNANN introduces a contrastive learning approach. At the user–genre level, the model makes users closer to genres they have under-explored, creating opportunities for discovery without pushing

them into completely unrelated areas. The key is that focusing on a single item often highlights very small, specific details, making the recommendation too narrow. By contrast, working at the genre level helps us push exploration across a wider boundary (a whole group of items) that the user hasn't seen, leading to discovery that feels more natural and meaningful. At the genre–item level, the model strengthens the alignment between genres and their items, so that when users are guided toward new genres, the recommended items are still coherent and meaningful. Together, these contrastive components ensure that novelty emerges as part of the models learned representation, rather than as a surface adjustment afterward.

By combining BPR for accuracy and contrastive learning for novelty, BaNANN learns to provide recommendations that are both familiar and fresh. Users continue to receive items they are likely to enjoy, while gradually being exposed to content that expands their horizons. This integrated design moves beyond post-processing methods and creates a systematic trade-off between exploration and exploitation.

2 Objectives

This paper introduces BaNANN (Balancing Novelty and Accuracy with a Tripartite Graph Neural Network and Dual Contrastive Learning), a framework designed to balance the two conflicting goals of recommender systems: accuracy and novelty by integrating both goals directly into its training process.

1. A **tripartite user–genre–item graph** that makes genre-level reasoning explicit (inspired by extended tripartite graphs [2]).

2. An **accuracy objective (BPR loss)** that keeps rankings consistent with user preferences.

3. A **novelty objective (contrastive loss)** that systematically encourages exploration in under-explored genres and maintains semantic consistency at the item level.

3 Scope

Use MovieLens dataset and evaluate accuracy and novelty metrics compared with baseline models, Optional ablations.

4 Project Activities

The project will be conducted in sequential phases, aligned with research methodology and system development principles. The main activities are listed below:

1. **Literature Review**
   Study recommender systems, novelty/accuracy trade-offs. Finalize the problem statement and evaluation plan.

2. **System and Model Design**
   Design the tripartite user–genre–item schema; define training objectives (BPR for accuracy, dual contrastive for novelty). Select the backbone and attention strategy (PyTorch Geometric: `GATConv`/heterogeneous GAT).

3. **Implementation**
   Implement BaNANN in PyTorch Geometric with a GAT-based encoder for the tripartite graph; build data preprocessing and evaluation pipelines.

4. **Testing and Evaluation**
   Run experiments on MovieLens; compare to BPR/LightGCN and re-ranking baselines [1]. Report accuracy (Hit@K, NDCG, Precision, Recall, MRR) and novelty metrics.

5. **Documentation and Reporting**
   Prepare reports and the final project report according to academic guidelines.

Figure 1: Gantt Timeline (Aug 2025 – Apr 2026)

Activities	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**
Literature Review									
System & Model Design									
Implementation									
Testing & Evaluation									
Documentation									

5 Benefits

1. **For the researcher**

   - Gain practical experience in designing and implementing advanced Graph Neural Networks (GNNs).
   - Learn how to apply modern techniques like Contrastive Learning to balance competing objectives.
   - Develop skills in evaluating and analyzing the trade-off between recommendation accuracy and novelty.

2. **For business and society**

- It helps mitigate the filter bubble effect by providing users with more diverse and novel recommendations and can be applied to business platforms (e.g., e-commerce, streaming) to improve long-term user engagement.

6 Equipment

This project will be implemented using the student's own notebook for coding and documentation, and Google Colab for model training and experiments. Since Colab provides GPU/TPU resources and the dataset (MovieLens) is open-source, no additional budget is required.

7 Budget

1. SanDisk Ultra microSD 256GB 570 THB
2. Screen Repair 4815 THB

Total: 5385 THB

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8 Appendix

As we talk, the core challenge in recommender systems lies in balancing the conflicting objectives of accuracy and novelty. Our method addresses this conflict by integrates a tripartite graph architecture, BPR, and two-step contrastive learning for balanced objective optimization. Our framework is show in figure 2.

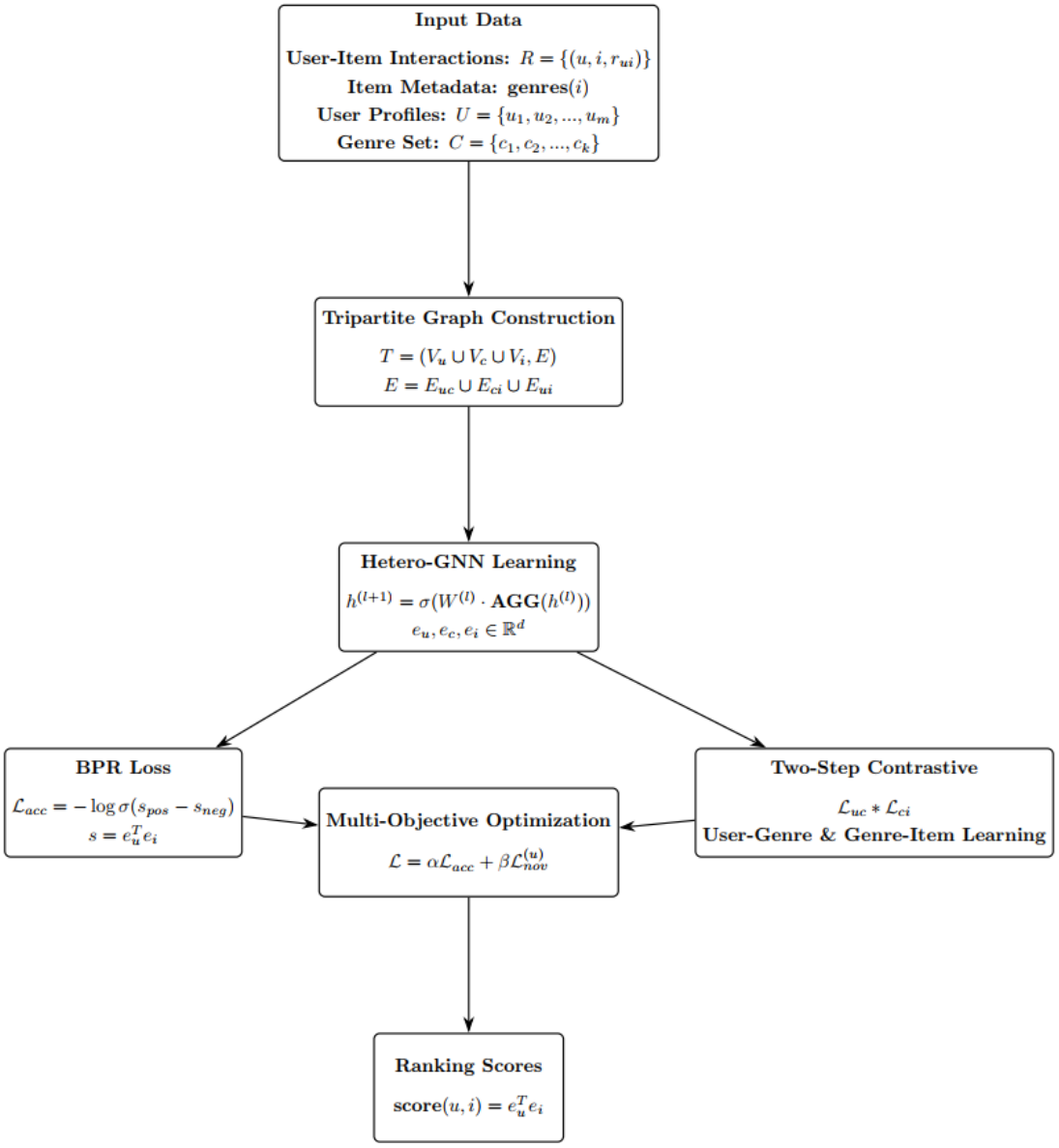


Figure 2: our model framework

8.1 Tripartite Graph Construction

Let $U = \{u_1, u_2, \ldots, u_m\}$ denote the set of m users, $I = \{i_1, i_2, \ldots, i_n\}$ the set of n items, and $C = \{c_1, c_2, \ldots, c_k\}$ the set of k genres. Observed user-item ratings are represented as $R = \{(u, i, r_{ui}) | u \in U, i \in I\}$, where r_{ui} denotes the ratings.

Tripartite Graph Construction

We propose a tripartite graph structure that explicitly incorporates content semantics through genre nodes, creating a three-layer architecture:

$$V = V_u \cup V_c \cup V_i \tag{1}$$

where $V_u = \{v_{u1}, \ldots, v_{um}\}$ are user nodes, $V_c = \{v_{c1}, \ldots, v_{ck}\}$ are genre nodes, and $V_i = \{v_{i1}, \ldots, v_{in}\}$ are item nodes.

The edge set E encompasses three distinct relationship types:

User-Item Edges (E_{ui}): Store direct collaborative filtering signals essential for accuracy. For each observed interaction $(u,i) \in R$, we create an edge if $r_{ui} \in R$.

Genre-Item Edges (E_{ci}): Encode content-based relationships that enable semantic reasoning. An edge $e_{ci} \in E_{ci}$ exists if item i belongs to genre c, based on item metadata.

User-Genre Edges (E_{uc}): Capture implicit user preferences at the content category level. For each user-genre pair (u,c), we first identify all items that user u has interacted with belonging to genre c:

$$I_{uc} = \{i \in I | (u,i) \in R \text{ and } (c,i) \in E_{ci}\} \tag{2}$$

The edge weight w_{uc} is computed as the average interaction strength across these items:

$$w_{uc} = \frac{1}{|I_{uc}|} \sum_{i \in I_{uc}} r_{ui} \tag{3}$$

Information Flow and Multi-hop Propagation

The tripartite structure enables rich information propagation patterns that enhance both accuracy and novelty:

One-hop Paths (User $\leftrightarrow$ Item/User $\leftrightarrow$ Genre): Preserve traditional collaborative filtering signals.

Two-hop Paths (User $\leftrightarrow$ Genre $\leftrightarrow$ Item/User $\leftrightarrow$ Item $\leftrightarrow$ Genre): Enable content-based reasoning and genre-level preference learning.

Note that our tripartite graph contains six directed edge types: $(user, rates, item)$ and $(item, ratesBy, user)$; $(user, likes, genre)$ and $(genre, likesBy, user)$; $(genre, has, item)$ and $(item, isIn, genre)$.

8.2 Heterogeneous Graph Neural Network Learning

Our tripartite graph contains multiple node types and edge types, requiring a heterogeneous graph neural network architecture. We employ HeteroConv layers that apply

different transformations to different edge types while enabling information flow across the entire graph.

For each edge type (s,r,t) where s and t are source and target node types and r is the relation type, we define a relation-specific convolution:

$$h_{t,v}^{(l+1)} = \sigma\left(W_r^{(l)} \cdot \mathrm{AGG}_{u \in N_r(v)}\left(h_{s,u}^{(l)}\right)\right) \tag{4}$$

where $h_{t,v}^{(l)}$ is the embedding of node v of type t at layer l, $N_r(v)$ represents neighbors of v connected via relation r, and $W_r^{(l)}$ is a weight matrix.

We implement the aggregation function using mean pooling:

$$\mathrm{AGG}_{u \in N_r(v)}\left(h_{s,u}^{(l)}\right) = \mathrm{MEAN}\left\{h_{s,u}^{(l)} : u \in N_r(v)\right\} \tag{5}$$

8.3 BPR Learning

We use a simple BPR loss show as follows:

$$s(u,i) = e_u^T e_i \tag{6}$$

For each user u, observed item i, and unobserved item j, the loss of BPR is the following.

$$\mathscr{L}_{\mathrm{acc}} = -\sum_{(u,i,j) \in D_s} \ln \sigma(s(u,i) - s(u,j)) \tag{7}$$

where D_s is the training set and $\sigma(\cdot)$ is the sigmoid function.

8.4 Two-Step Contrastive Learning Framework

User-Genre Contrastive Learning

For each user u, we analyze their interaction pattern across genres to compute a preference distribution. Let n_{uc} denote the number of interactions user u has had with items in genre g:

$$n_{uc} = \sum_{i \in I:(c,i) \in E_{ci}} \mathbb{I}[(u,i) \in R] \tag{8}$$

where the indicator function $\mathbb{I}[A]$ is defined as:

$$\mathbb{I}[A] = \begin{cases} 1 & \text{if condition } A \text{ is true} \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

The probability that user u interacts with genre c is:

$$p_{uc} = \frac{n_{uc}}{\sum_{c' \in C} n_{uc'}} \tag{10}$$

The first step is to encourage users to be attracted to genres they haven't sufficiently explored. So, for each user u, we identify unexplored genres as positive samples. A genre c is considered unexplored for user u if $p_{uc} < \varepsilon$ where $\varepsilon = 1/|C|$ represents uniform exploration.

Let C_u^- denote the set of unexplored genres for user u:

$$C_u^- = \{c \in C | p_{uc} < \varepsilon\} \tag{11}$$

The user-genre contrastive loss encourages user embeddings to be similar to unexplored genre embeddings:

$$\mathscr{L}_{uc} = -\sum_{u \in U} \sum_{c \in C_u^-} \ln \frac{\exp(e_u^T e_c / \tau)}{\sum_{c' \in C} \exp(e_u^T e_{c'} / \tau)} \tag{12}$$

Genre-Item Contrastive Learning

The second step ensures that when users are attracted to new genres, the recommended items within those genres are semantically appropriate. For each genre c, we want item embeddings within that genre to be similar to the genre embedding.

Let I_c denote the set of items belonging to genre c:

$$I_c = \{i \in I | (c,i) \in E_{ci}\} \tag{13}$$

The genre-item contrastive loss is:

$$\mathscr{L}_{ci} = -\sum_{c \in C} \sum_{i \in I_c} \ln \frac{\exp(e_c^T e_i / \tau)}{\sum_{i' \in I} \exp(e_c^T e_{i'} / \tau)} \tag{14}$$

8.5 Multi-Objective Optimization

The overall novelty loss combines both contrastive components:

$$\mathscr{L}_{\text{nov}} = \mathscr{L}_{uc} \mathscr{L}_{ci} \tag{15}$$

Then we combine BPR loss and contrastive loss into the final loss function as follows:

$$\mathscr{L} = \alpha \mathscr{L}_{\text{acc}} + \beta \mathscr{L}_{\text{nov}} \tag{16}$$

8.6 Inference and Recommendation Generation

During inference, we compute recommendation scores using the simple scoring function used during training:

$$\text{score}(u, i) = e_u^T e_i \tag{17}$$

Our task is to perform ranking. After this score is computed for all candidate items i for a given user u, we generate the final top-K recommendation list by ranking these items in descending order of their scores.

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